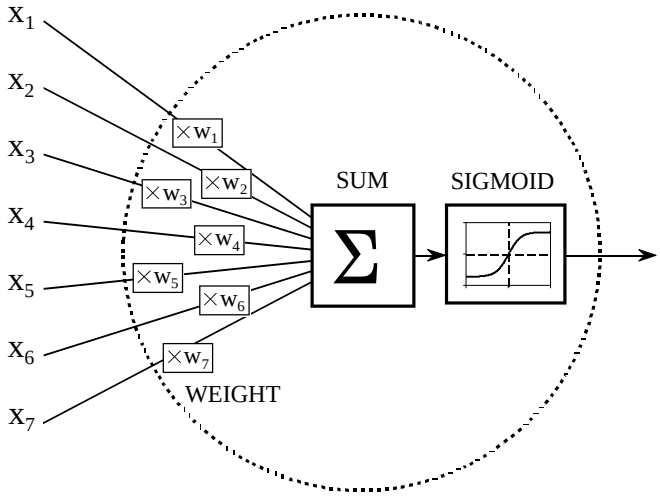

FIGURE 26-6
Neural network active node. This is a flow diagram of the active nodes used in the hidden and output layers of the neural network. Each input is multiplied by a weight (the w_N values), and then summed. This produces a single value that is passed through an "s" shaped nonlinear function called a *sigmoid*. The sigmoid function is shown in more detail in Fig. 26-7.

With other weights, the outputs might classify the objects as: metal or non-metal, biological or nonbiological, enemy or ally, etc. No algorithms, no rules, no procedures; only a relationship between the input and output dictated by the values of the weights selected.

```
100 'NEURAL NETWORK (FOR THE FLOW DIAGRAM IN FIG. 26-5)
110 '
120 DIM X1[15]                      'holds the input values
130 DIM X2[4]                       'holds the values exiting the hidden layer
140 DIM X3[2]                       'holds the values exiting the output layer
150 DIM WH[4,15]                    'holds the hidden layer weights
160 DIM WO[2,4]                     'holds the output layer weights
170 '
180 GOSUB XXXX                      'mythical subroutine to load X1[ ] with the input data
190 GOSUB XXXX                      'mythical subroutine to load the weights, WH[ , ] & W0[ , ]
200 '
210 '                               'FIND THE HIDDEN NODE VALUES, X2[  ]
220 FOR J% = 1 TO 4                 'loop for each hidden layer node
230   ACC = 0                       'clear the accumulator variable, ACC
240   FOR I% = 1 TO 15              'weight and sum each input node
250     ACC = ACC + X1[I%] * WH[J%,I%]
260   NEXT I%
270   X2[J%] = 1 / (1 + EXP(-ACC) )     'pass summed value through the sigmoid
280 NEXT J%
290 '
300 '                               'FIND THE OUTPUT NODE VALUES, X3[  ]
310 FOR J% = 1 TO 2                 'loop for each output layer node
320   ACC = 0                       'clear the accumulator variable, ACC
330   FOR I% = 1 TO 4               'weight and sum each hidden node
340     ACC = ACC + X2[I%] * WO[J%,I%]
350   NEXT I%
360   X3[J%] = 1 / (1 + EXP(-ACC) )     'pass summed value through the sigmoid
370 NEXT J%
380 '
390 END
```

TABLE 26-1